Date:-

(1) $\int \frac{1}{x^2 - a^2} \cdot dx$

$= \frac{1}{(x-a)(x+a)}$

$= \frac{A}{(x-a)} + \frac{B}{(x+a)}$

$\therefore$ Solve for A and B,

$1 = A(x+a) + B(x-a)$

let, $x = a$, then $1 = A(2a) + B(0)$

$\therefore A = \frac{1}{2a}$

let, $x = -a$, then $1 = A(0) + B(-2a)$

$\therefore B = \frac{-1}{2a}$

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$$\int \frac{1}{x^2-a^2}dx = \int\left(\frac{1}{2a(x-a)} - \frac{1}{2a(x+a)}\right)dx$$

$$= \frac{1}{2a}\int\left(\frac{1}{(x-a)} - \frac{1}{(x+a)}\right)dx$$

$$= \frac{1}{2a}\left(\log|x-a| - \log|x+a|\right) + C$$

$$\boxed{= \frac{1}{2a}\log\left|\frac{x-a}{x+a}\right| + C}$$

(2) $\int \frac{1}{\sqrt{a^2-x^2}}$

$\therefore$ It is standard form.

$$\boxed{\text{Ans} = \sin^{-1}\left(\frac{x}{a}\right) + C}$$

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(3) $\dfrac{\cos 3x - \cos 4x}{\sin 3x + \sin 4x}$

Let $u = \sin 3x + \sin 4x$

$$\frac{du}{dx} = 3\cos 3x + 4\cos 4x$$

Now,

$$\Rightarrow \cos 3x - \cos 4x = -2\sin\frac{7x}{2}\sin\frac{-x}{2}$$

$$= 2\sin\frac{7x}{2}\cdot\sin\frac{x}{2}$$

$$\Rightarrow \sin 3x + \sin 4x = 2\sin\frac{7x}{2}\cos\frac{x}{2}$$

$$\int \frac{2\sin\frac{7x}{2}\sin\frac{x}{2}}{2\sin\frac{7x}{2}\cos\frac{x}{2}}dx = \int \frac{\sin\frac{x}{2}}{\cos\frac{x}{2}}dx$$

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$$= \int \tan \frac{x}{2} \, dx.$$

let $v = \frac{x}{2}$, $dv = \frac{1}{2} dx$

, $dx = 2dv$

$$\int \tan \frac{x}{2} \, dx = \int \tan v . 2dv$$

$$= 2 \int \tan v . dv$$

$$= 2 \int \frac{\sin v}{\cos v} . dv$$

Here, $w = \cos v$, so $dw = -\sin v \, dv$

$$2 \int \frac{\sin v}{\cos v} dv = -2 \int \frac{dw}{w}$$

$$= -2 \ln |w| + c$$

$$= -2 \ln |\cos v| + c$$

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~~Ad~~ And $v = \frac{x}{2}$

$$\boxed{\text{Ans} = -2\ln\left|\cos\frac{x}{2}\right| + C}$$